

YOLOv8 Technical Report

Equipment Detection - Controlled Laboratory Dataset

1. Executive Summary

This technical report presents the evaluation metrics for a YOLOv8 model tailored for Equipment Detection. The model was trained over 20 epochs using an exclusively controlled laboratory dataset. By utilizing this strictly controlled environment, varying lighting conditions and complex background clutter were minimized.

0.864

PRECISION

0.838

RECALL

0.892

MAP@50

2. External Resources

The dataset and the trained model weights are stored securely. You can access them via the following links:

Laboratory Dataset Link :

https://drive.google.com/file/d/1f2mvXx-DxgNv3ATM5QliAlte7gfubT_t/view?usp=drive_link

Trained Model (.pt) Link :

https://drive.google.com/file/d/1ZzlODsPwqsHFUQftiQpgnAcZ1_Fg0nYI/view?usp=drive_link

3. Training and Loss Trajectories

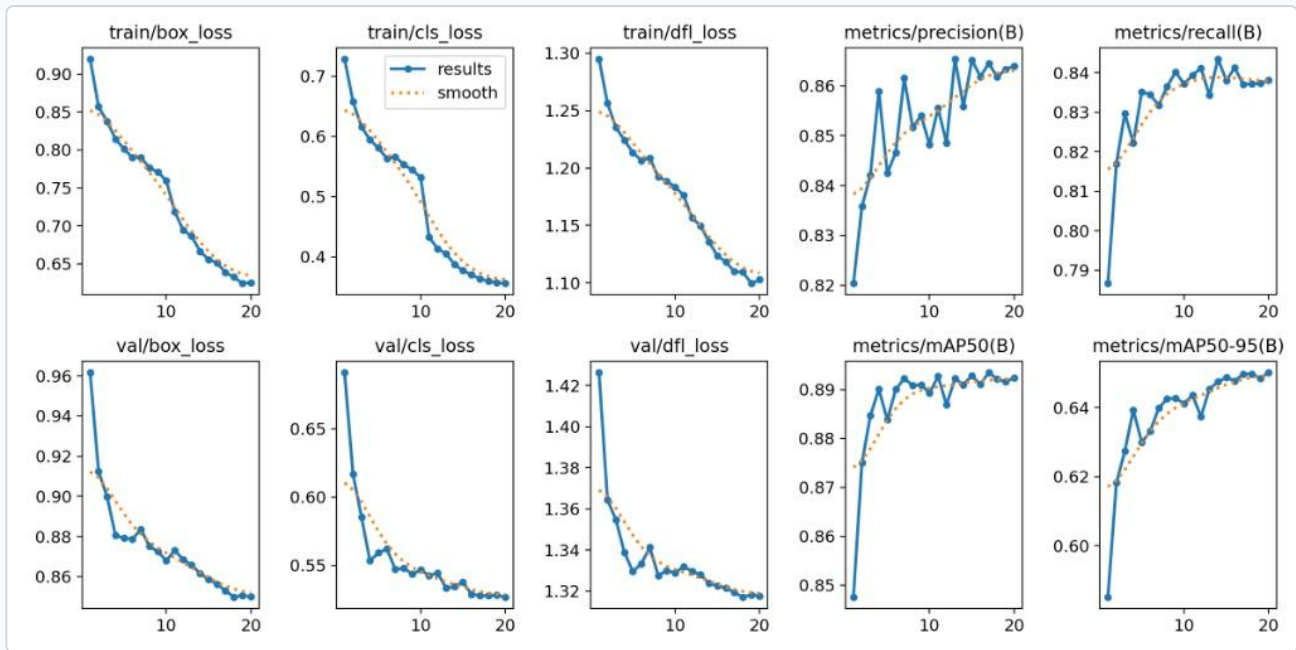


Figure 1: Comprehensive overview of loss functions and validation metrics.

4. Confidence and Validation Metrics

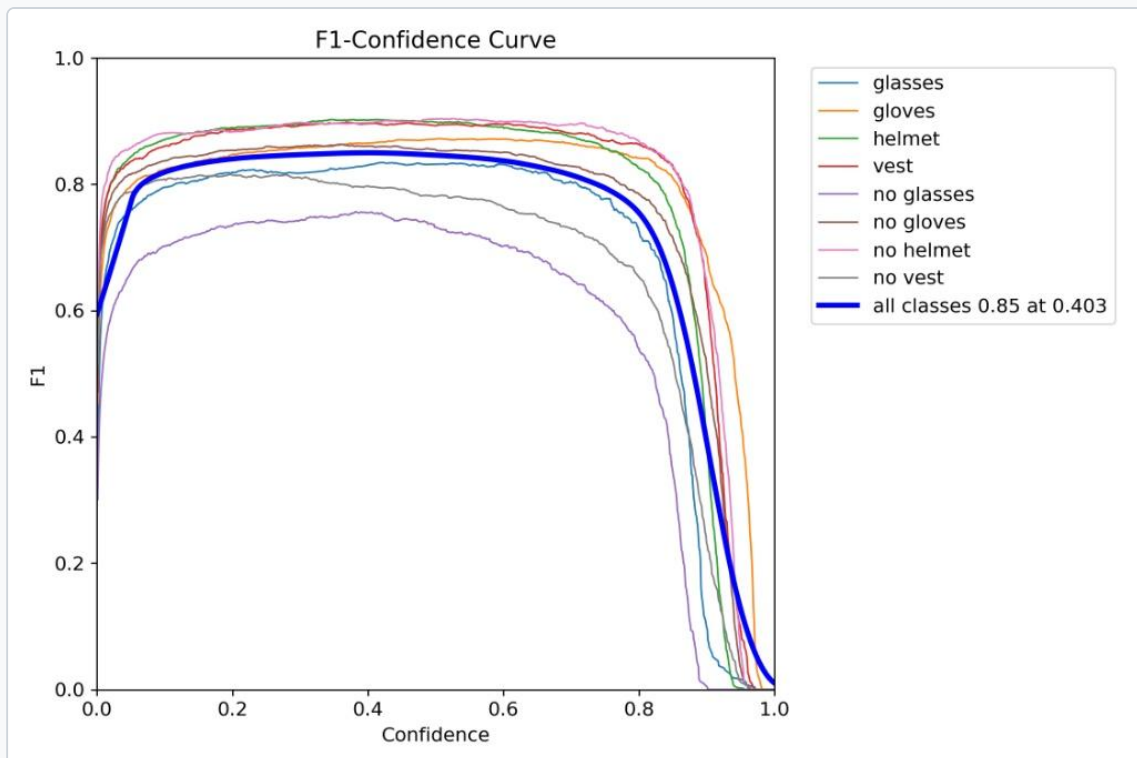


Figure 2: F1-Confidence Curve highlighting the optimal decision threshold.

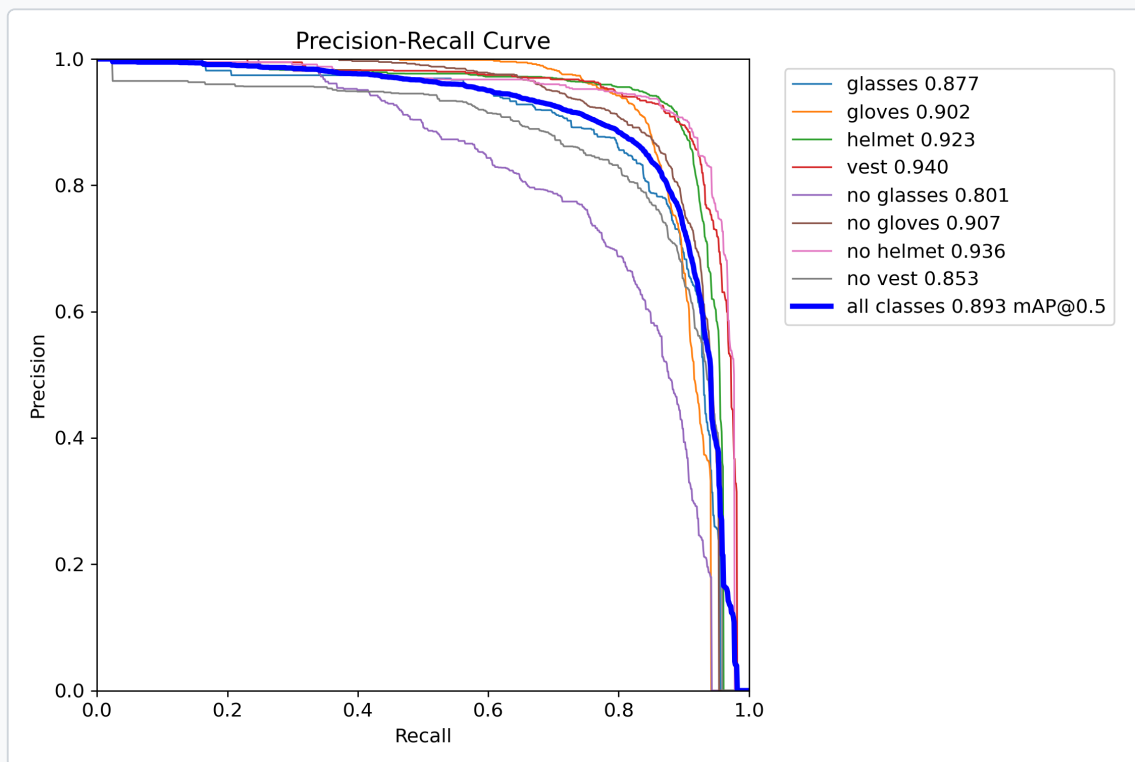


Figure 3: Precision-Recall Curve showing the trade-off characteristics.

5. Confusion Matrix

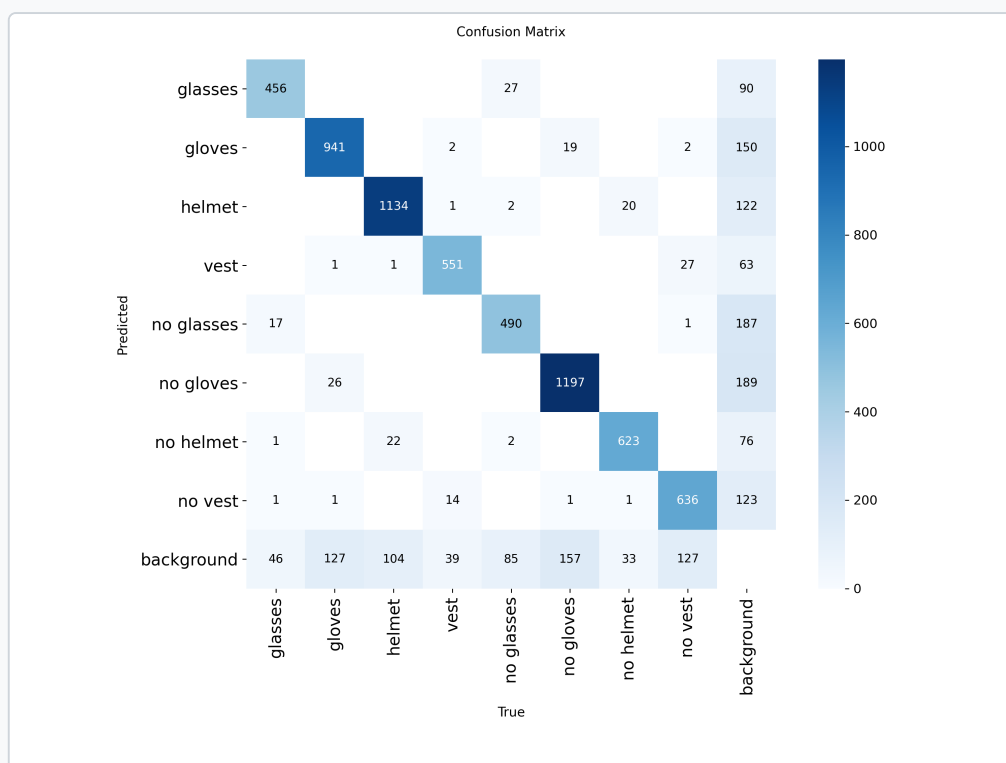


Figure 4: Validation Confusion Matrix for the equipment classes.

6. Dataset Morphological Distribution

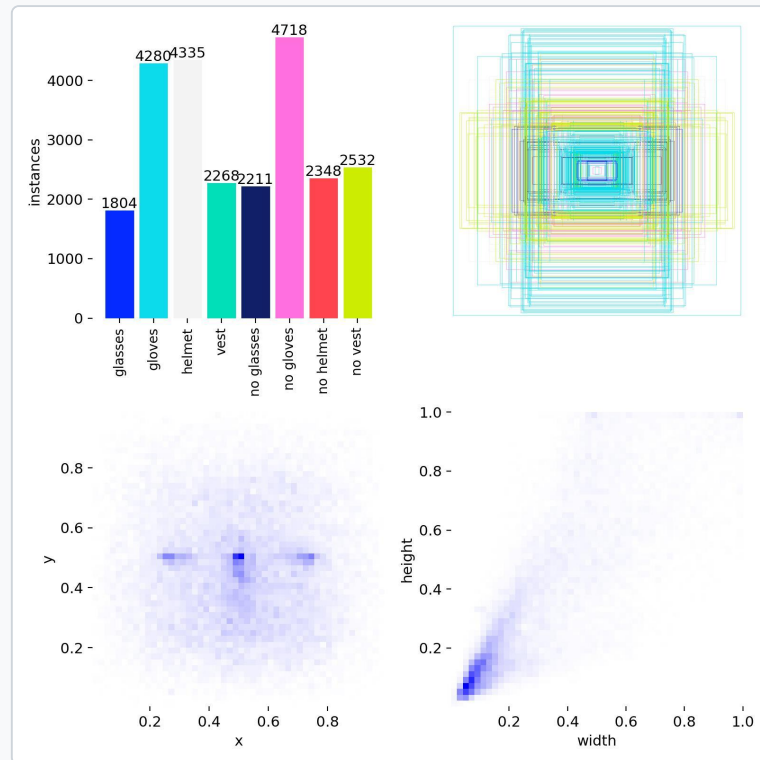


Figure 5: Bounding box dimension and spatial distribution mapping.